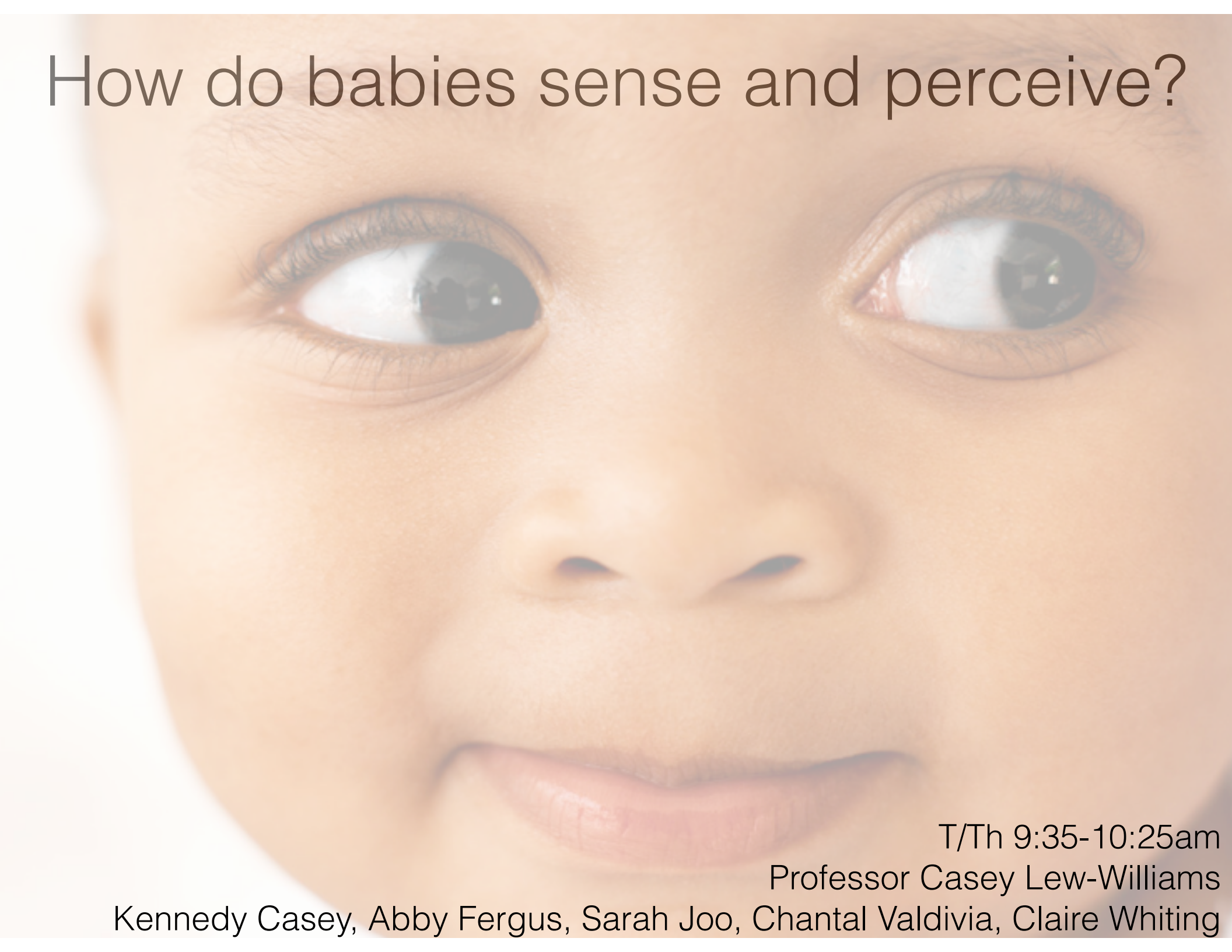


How do babies sense and perceive?

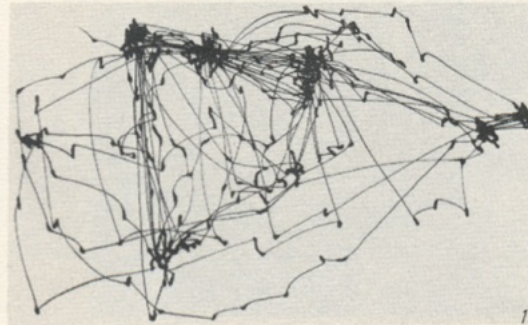


T/Th 9:35-10:25am

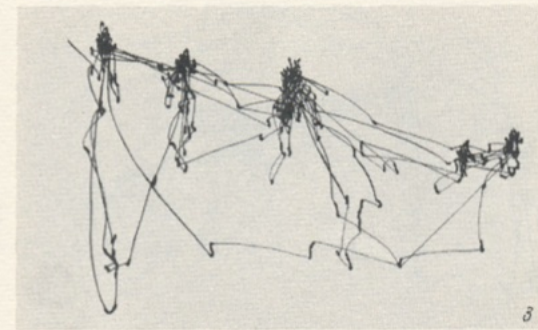
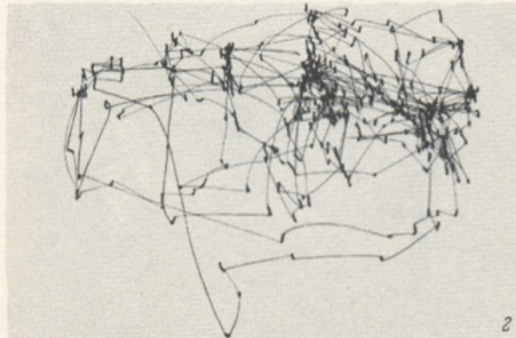
Professor Casey Lew-Williams

Kennedy Casey, Abby Fergus, Sarah Joo, Chantal Valdivia, Claire Whiting

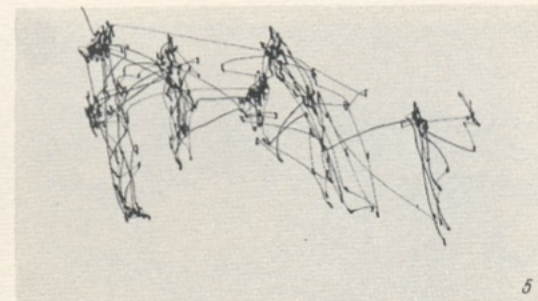
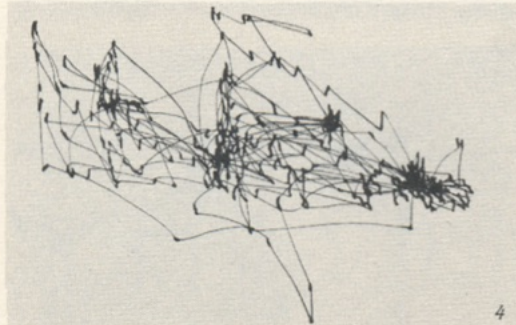




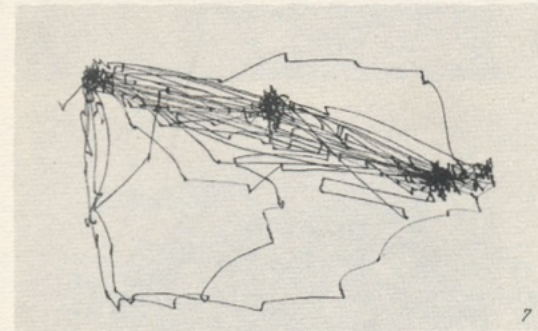
Free examination
of the picture



Give the ages of
the people



Remember the
clothes worn by
the people



Estimate how
long the visitor
had been away
from the family

Estimate material
circumstances of
the family in the
picture

What was the
family doing
before arrival of
the visitor?

Remember
position of
people & objects
in the room

Fetal sensory capabilities



- Taste and smell
 - Taste buds form at ~11 weeks
 - Fetus swallows amniotic fluid
 - Newborns recognize smell of own fluid
- Vision
 - At 26 weeks: bright light —> changes ❤️ rate
- Touch
 - Sensory receptors are strong enough to sense and respond to touch ~6-8 weeks of gestation

Early thinking about perception

- **William James**

- Perceptual abilities must be learned

- **Eleanor & J.J. Gibson**

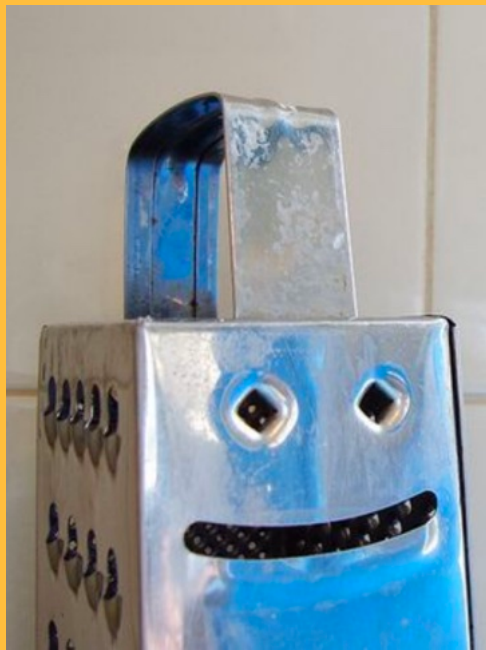
- Humans evolved in a world of objects and events, so some perception must be built in
- Perception is essential for survival
- But learning enhances the ability to detect and interpret information in the environment



Today

- Fetal sensory capabilities
- Visual development
 - What do babies see?
 - Effects of experience: Cataracts
 - Face processing
 - Depth perception
- Auditory development
 - What do babies hear?
 - Auditory development with cochlear implants
- Intermodal perception
 - McGurk effect

Visual development



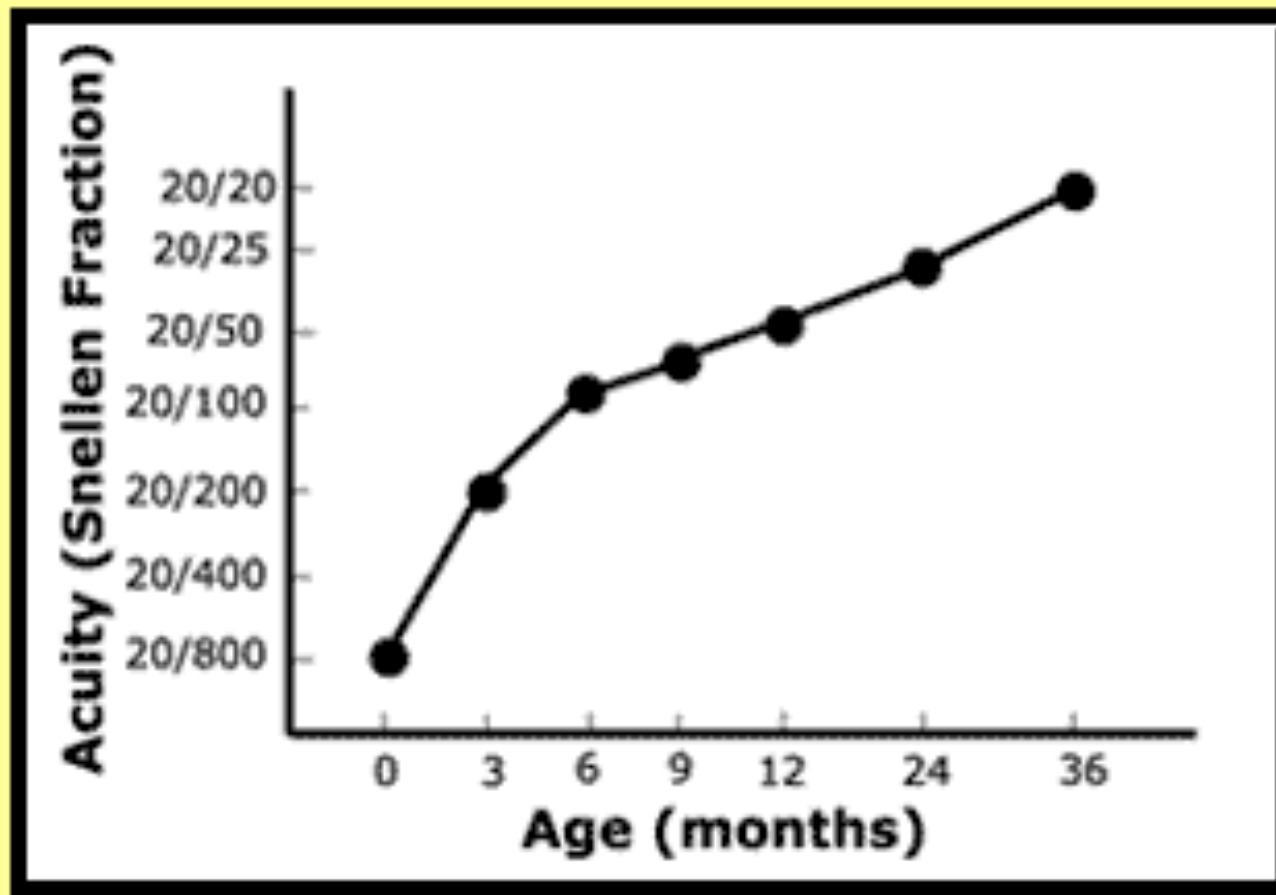
Thatcher Effect



What do newborns see?

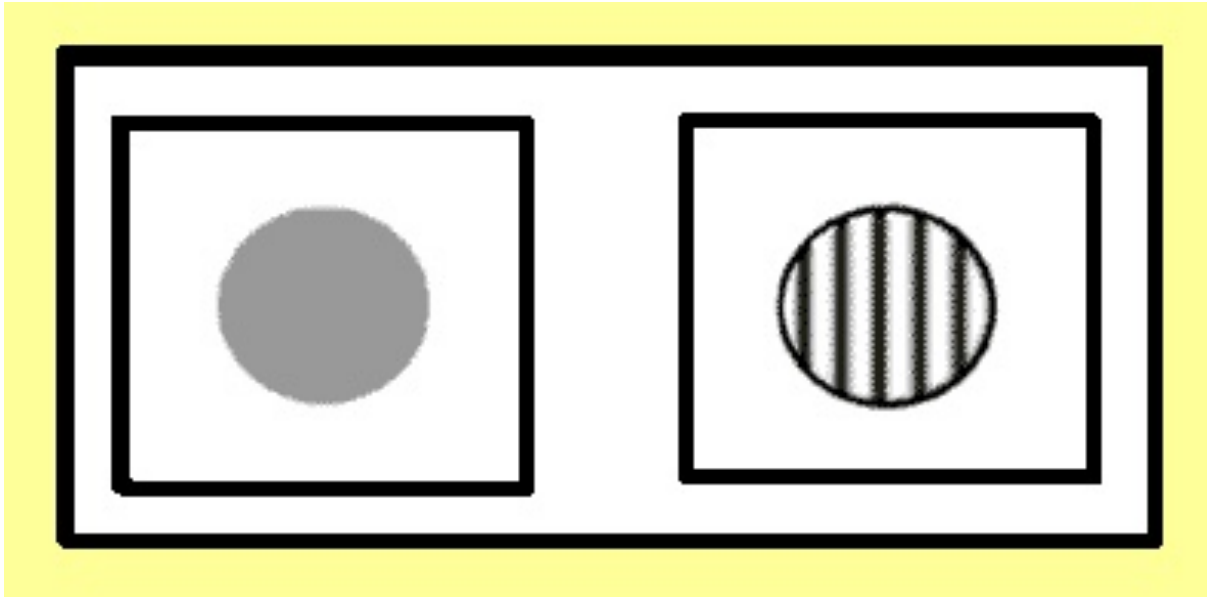


What do babies see?



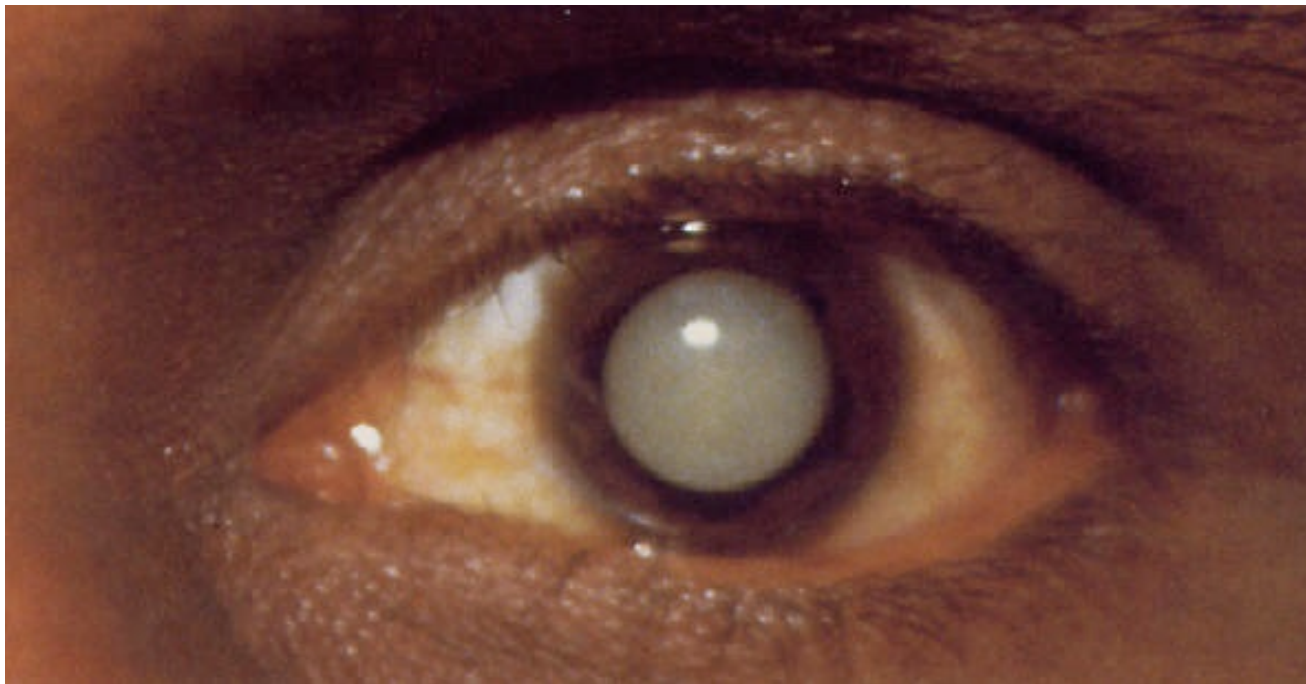
What do babies see?

- How do we know what baby vision looks like?

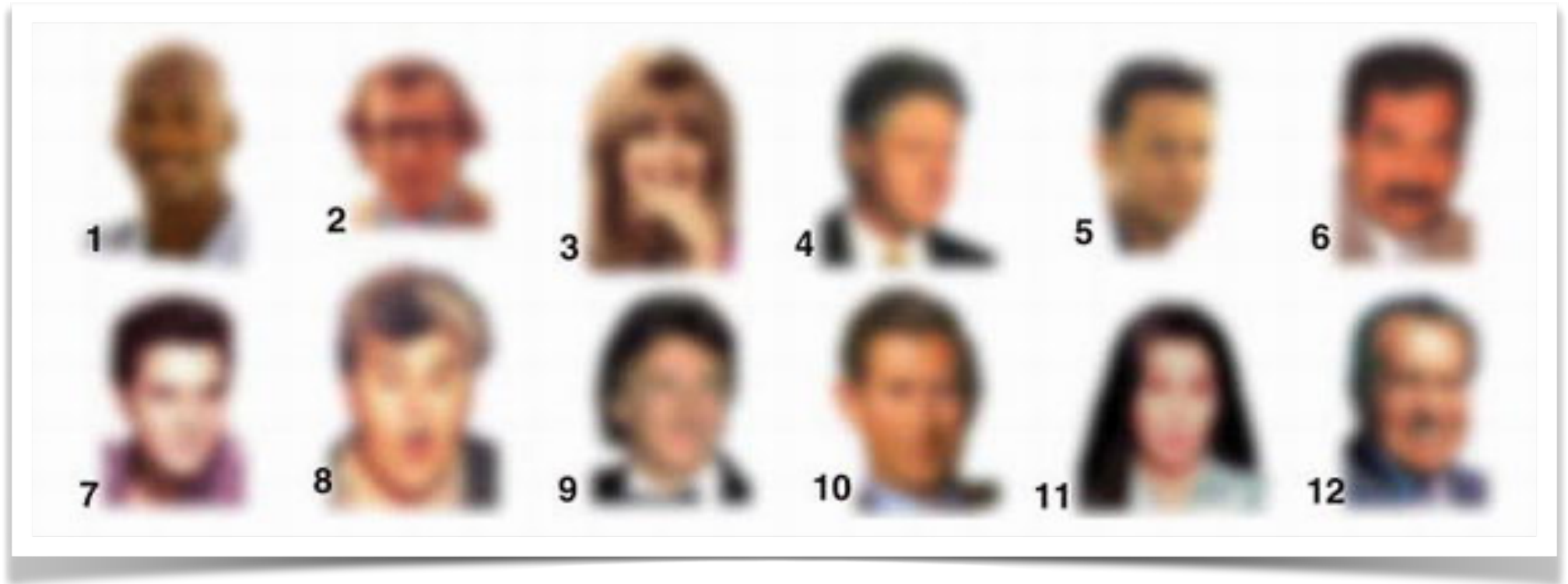


Effects of experience: Cataracts

- Cataract: clouding of the lens inside the eye
- Why is a faulty lens a bigger problem for infants vs. adults?
- How quickly does vision improve after surgery?
- Why patch the good eye after surgery?



Face processing



Face processing

- Faces: the most frequent pattern in the visual environment!



Face processing in utero?

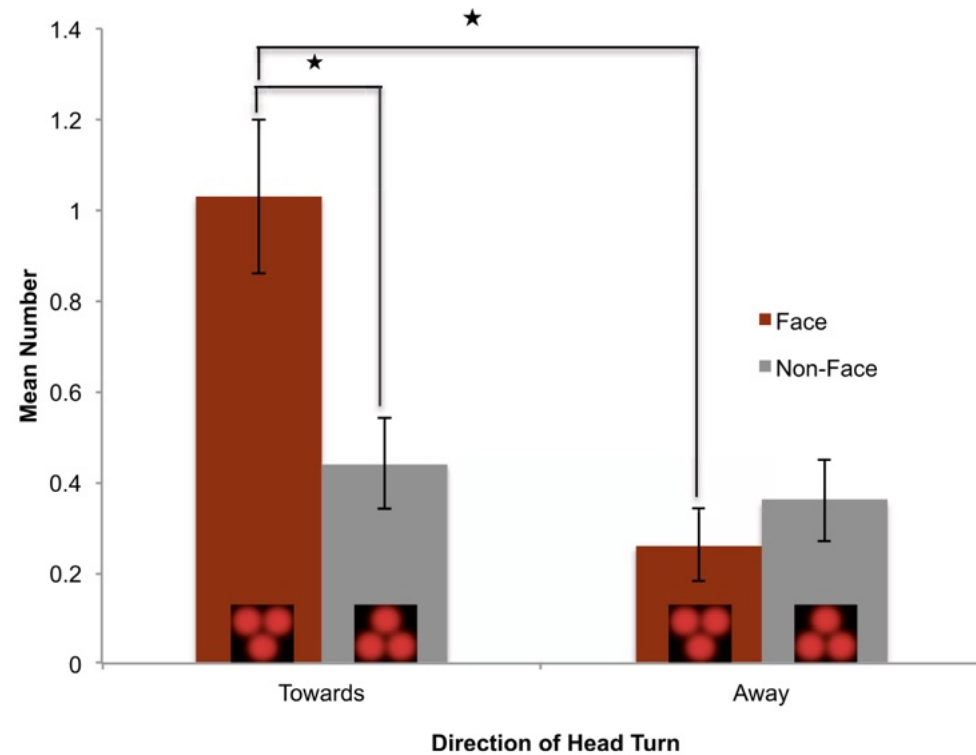
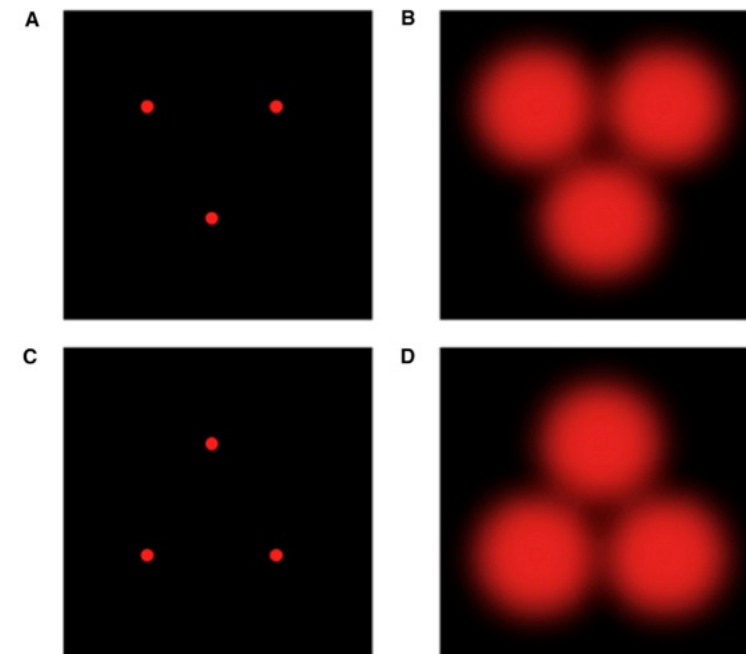


Figure 2. The Mean Number of Fetal Head Turns to the Stimuli

The mean number of head turns made toward (left two bars) and away (right two bars) for face-like (red) and non-face-like (gray) stimuli. Error bars represent standard error.

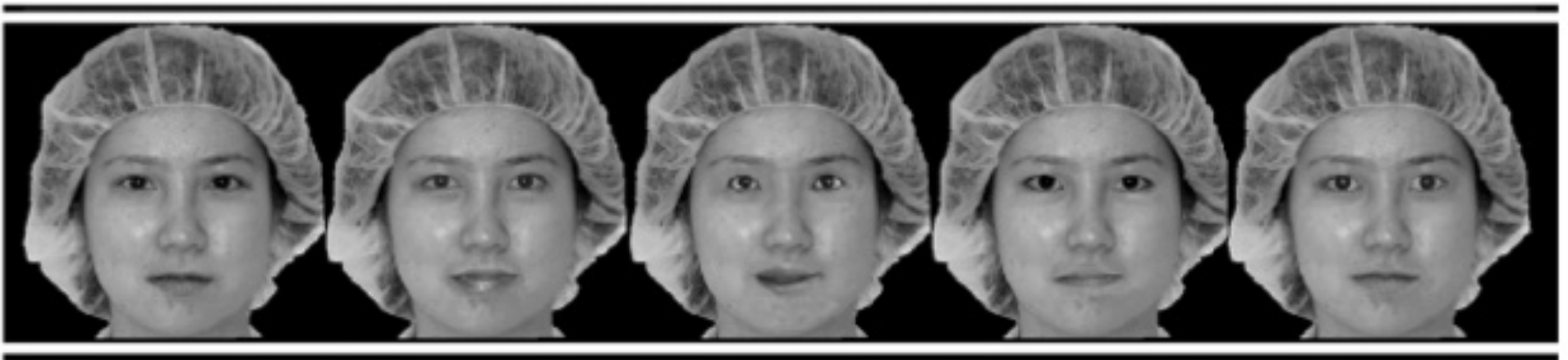
Face processing

- Faces: the most frequent pattern in the visual environment!



Face processing in **autism spectrum disorder (ASD)**

- Impairments in social interaction and language; restricted behaviors and interests
 - Currently ~1 in 68 children, boys > girls 5:1
 - Large range of functioning
- Children and adults with ASD perform poorly on face recognition tasks



Autistic Group

Control Group

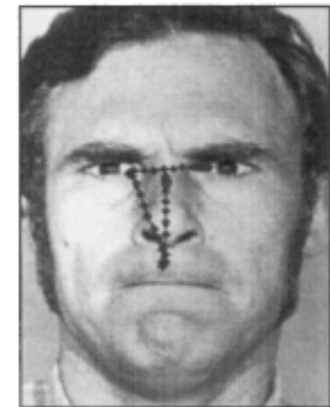
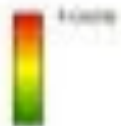
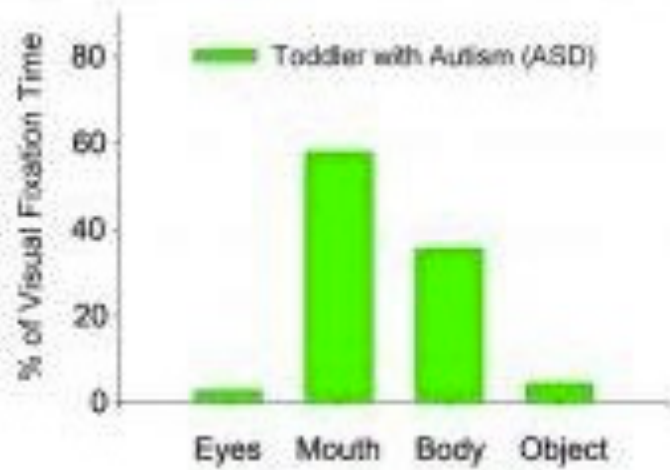
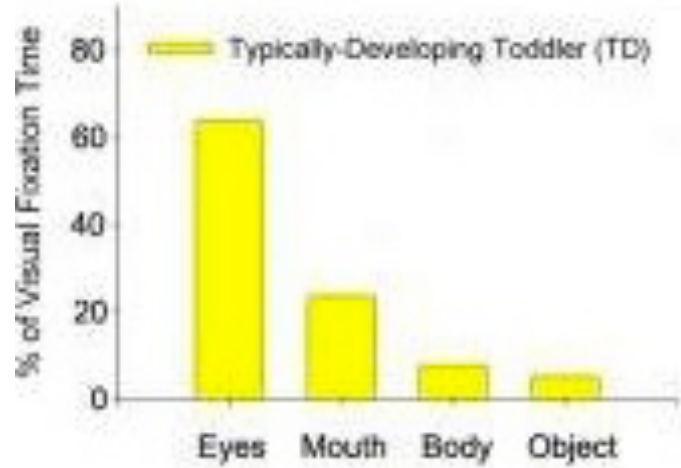
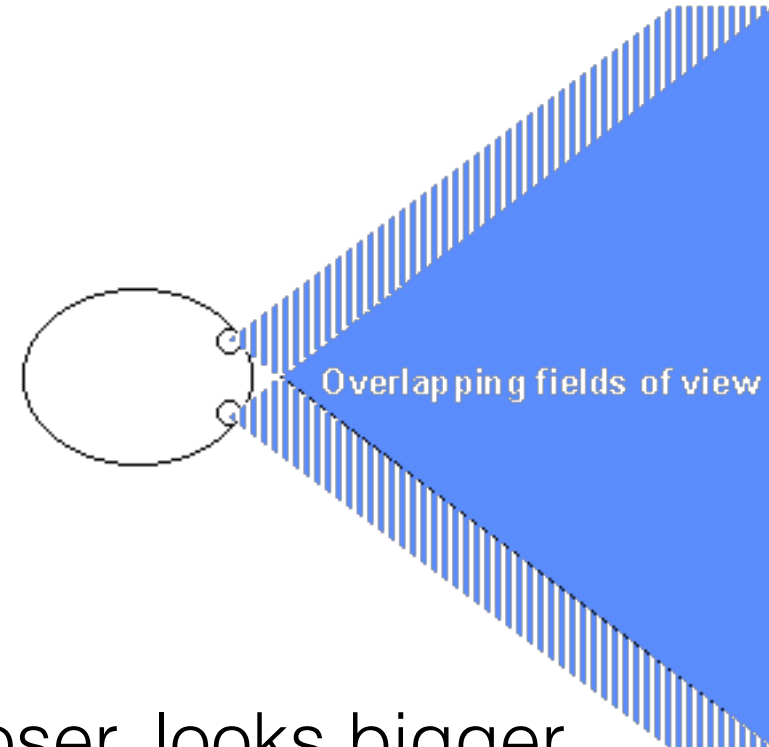


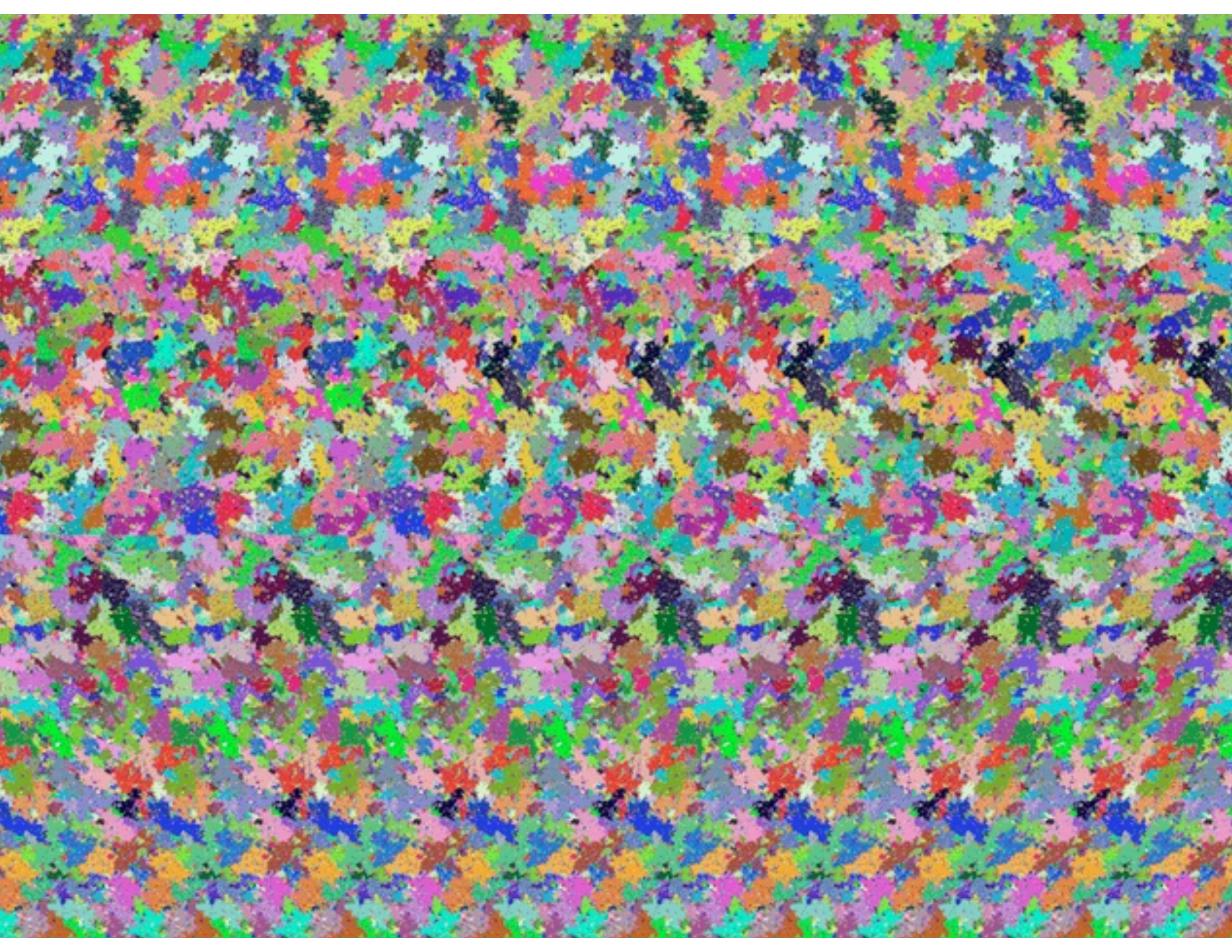
Fig. 1. Sample scanpaths from phase I of the experiment for three autistic participants (first column) and three control participants (second column). Participants were instructed to examine the faces in any manner they selected.



Depth perception



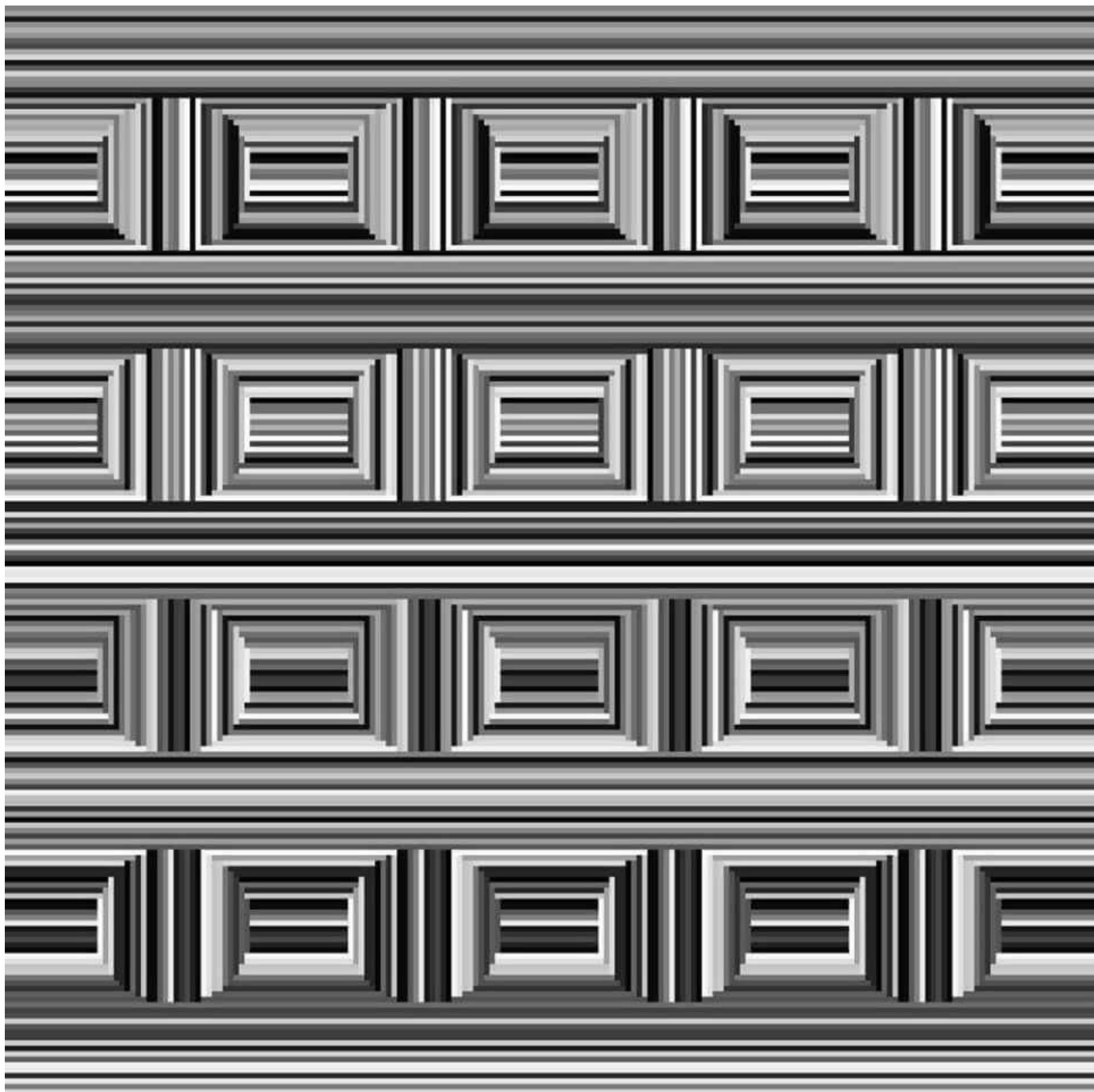
- **Optical expansion:** object comes closer, looks bigger
- **Binocular cues:** our 2 eyes have overlapping fields of vision. The disparity between these fields helps you locate things.
- **Monocular cues / Pictorial depth cues**







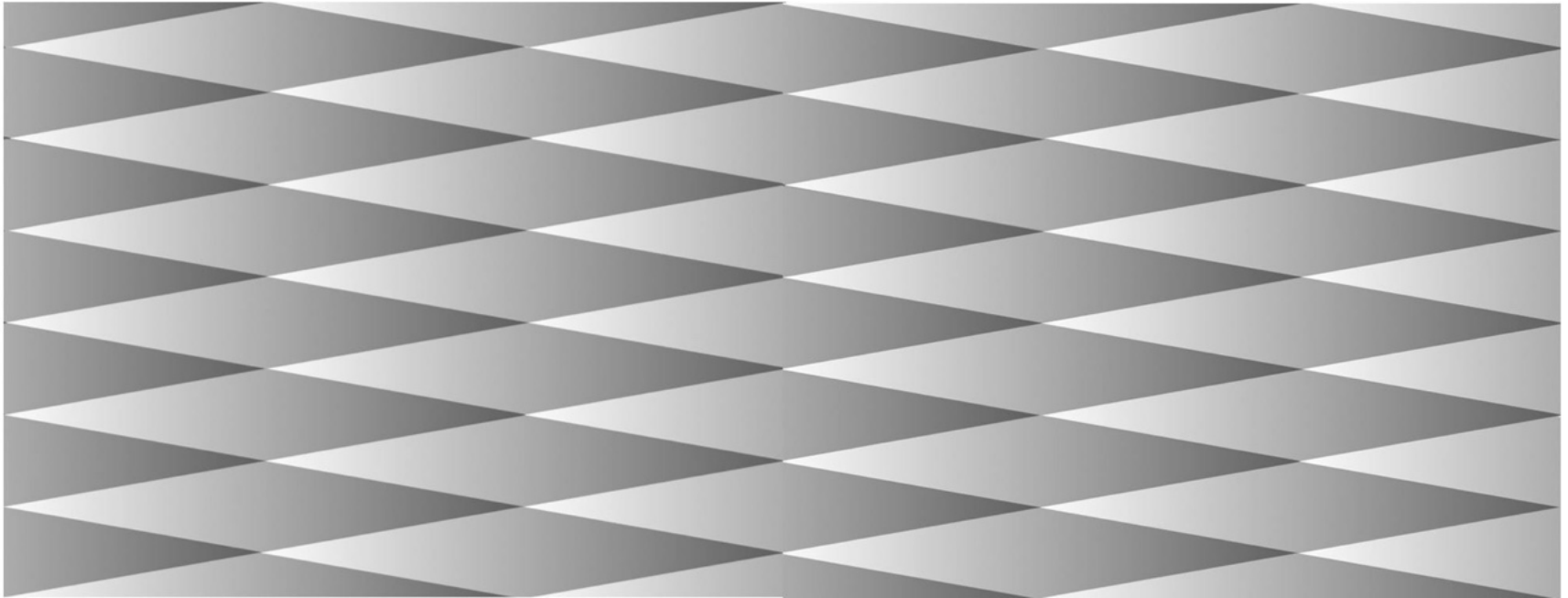




You're about to see the waiting
room of a science laboratory.

See if you can spot the items in the
room that are a little “out of place.”





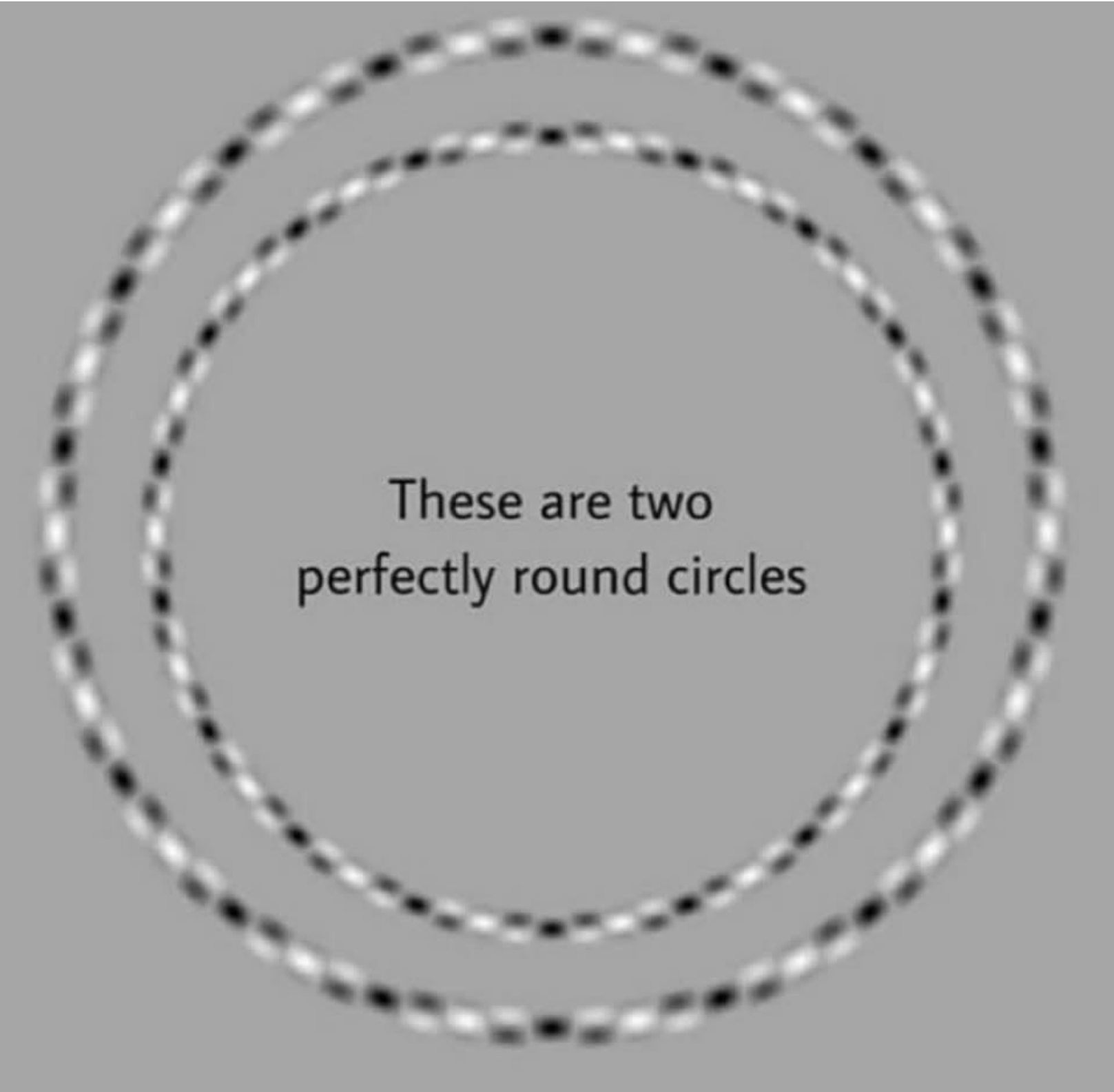




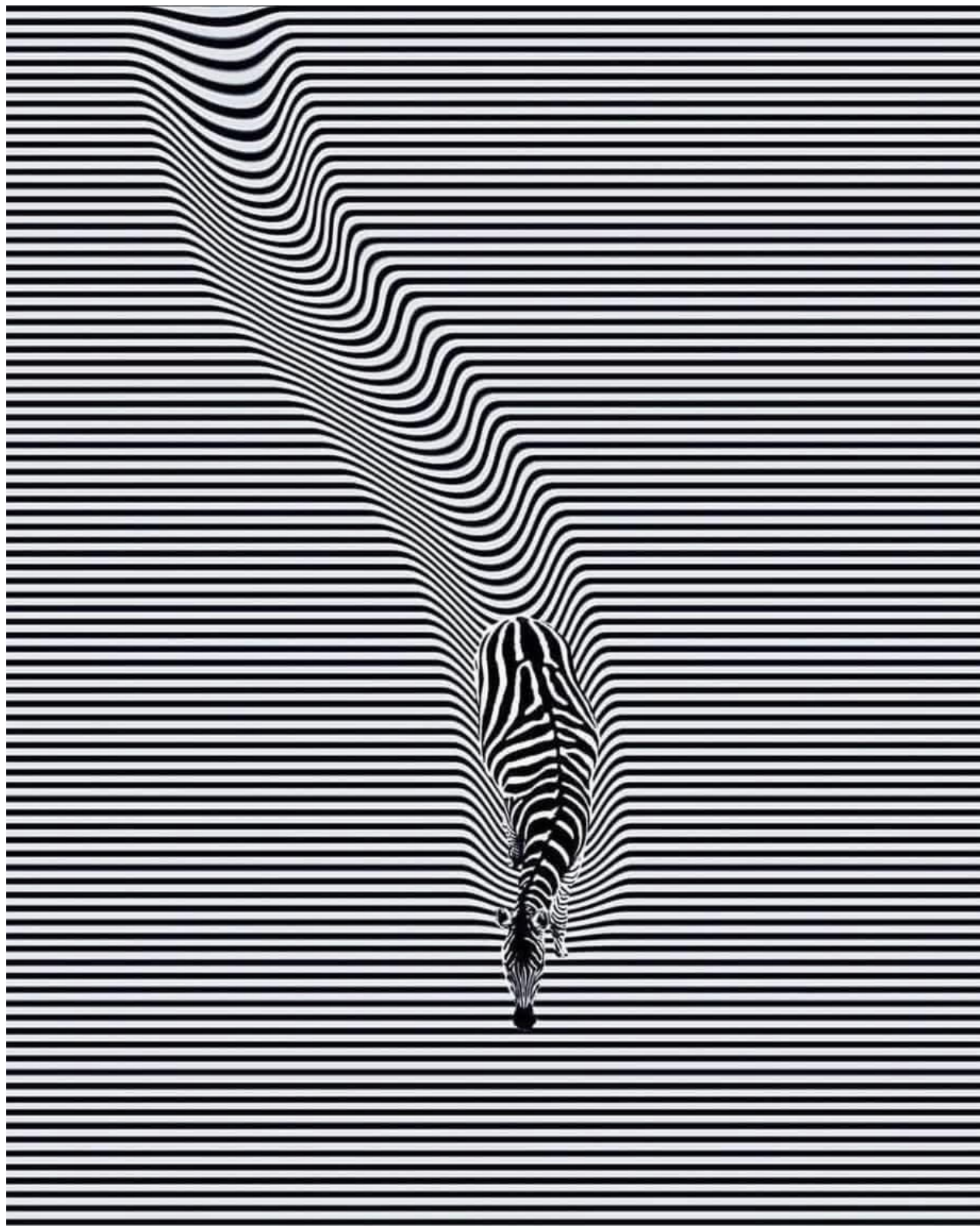
I was checking out this goat, dog, pig,
hybrid Chupacabra looking animal thing
for way too long before I realized it...

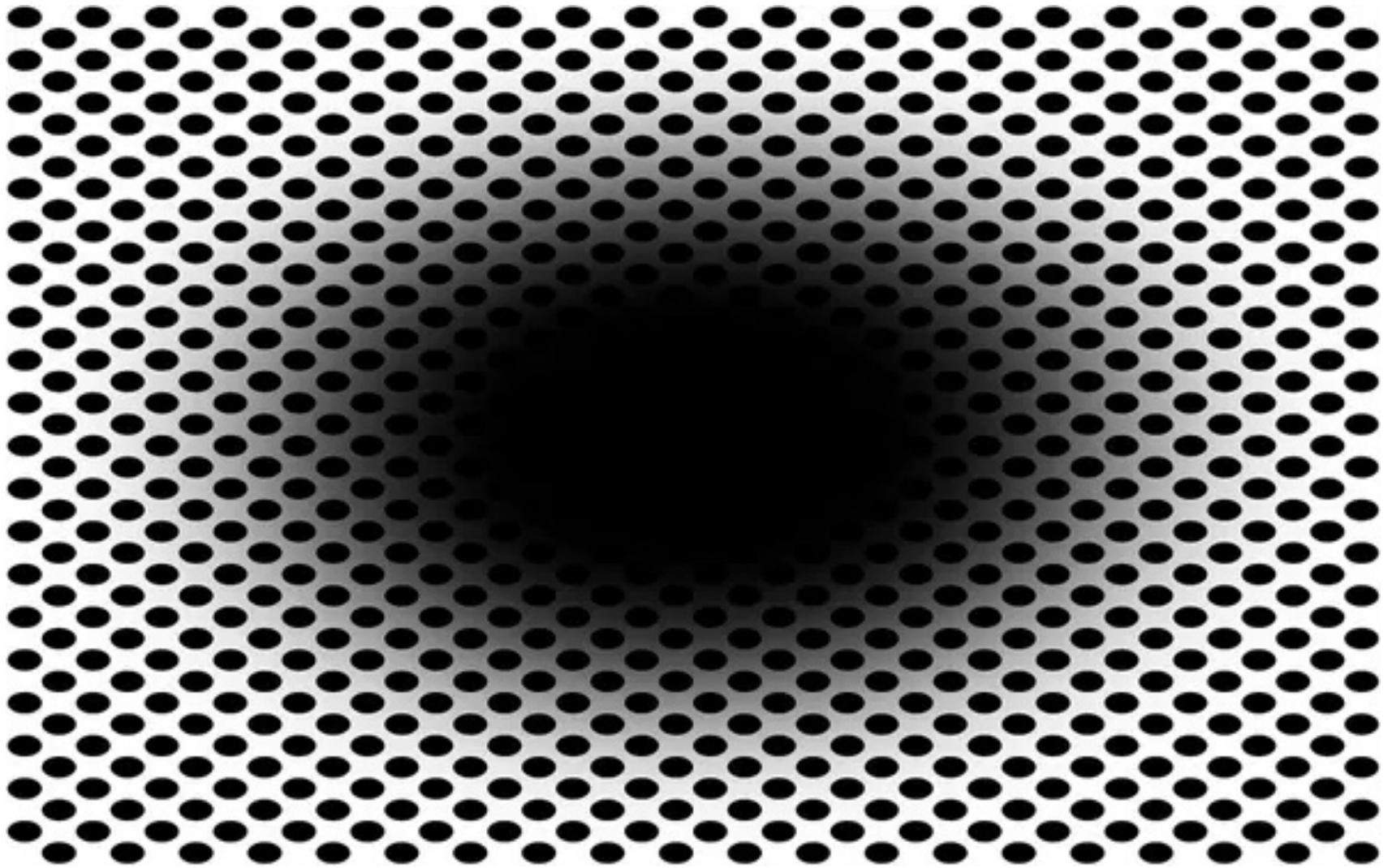


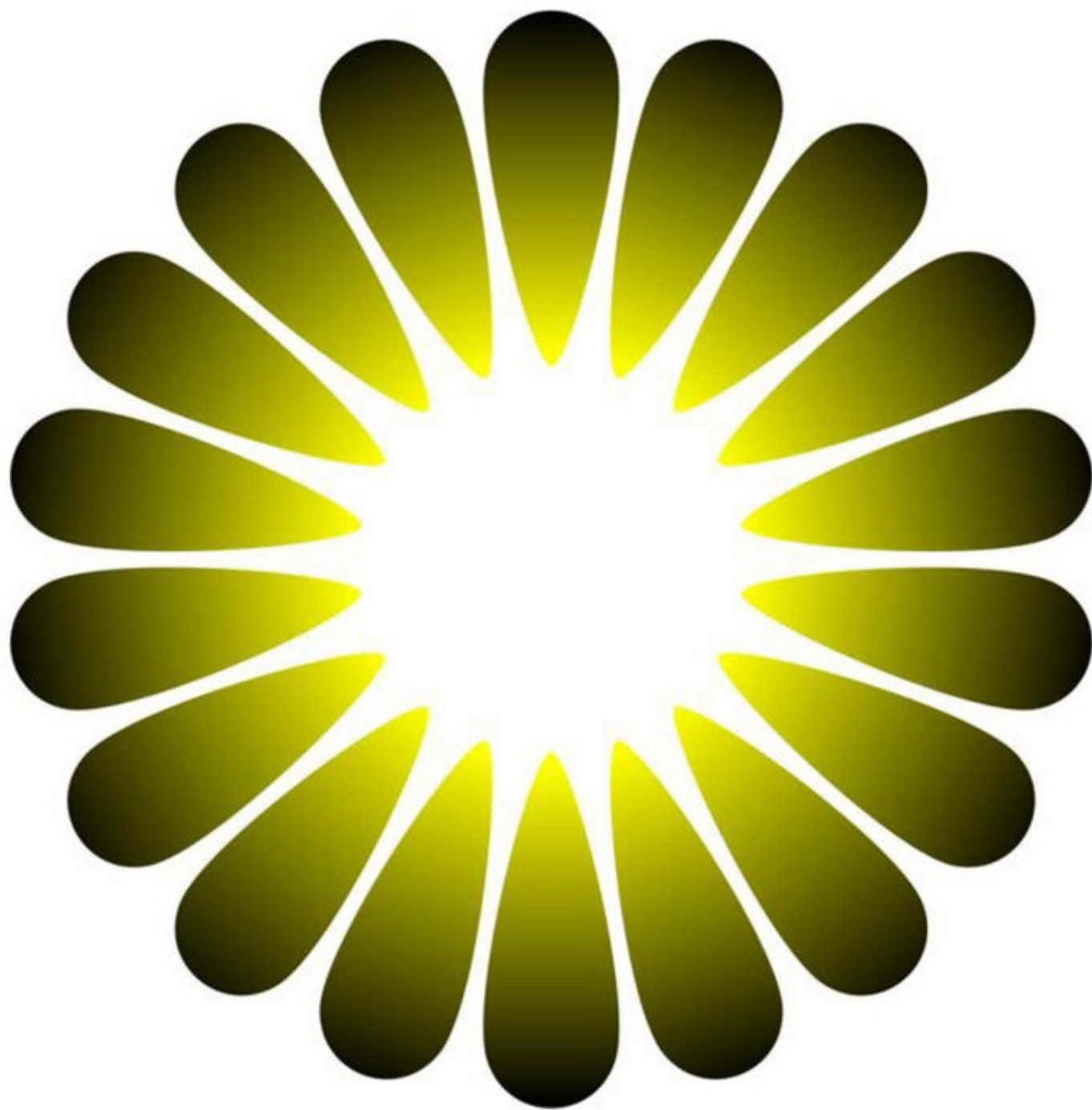


The image features two concentric circles drawn with a dashed line pattern on a solid gray background. The circles are centered and perfectly round. In the center of the circles, the text "These are two perfectly round circles" is written in a black, sans-serif font.

These are two
perfectly round circles







Picture a blinding sunny sky

Picture a dark room

Assessing depth perception

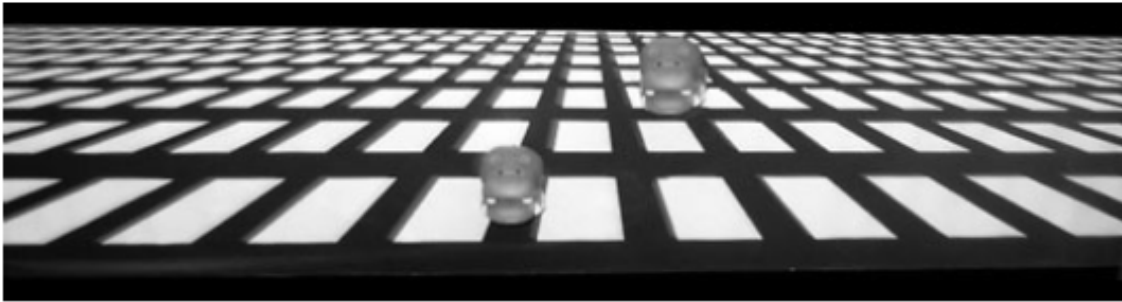
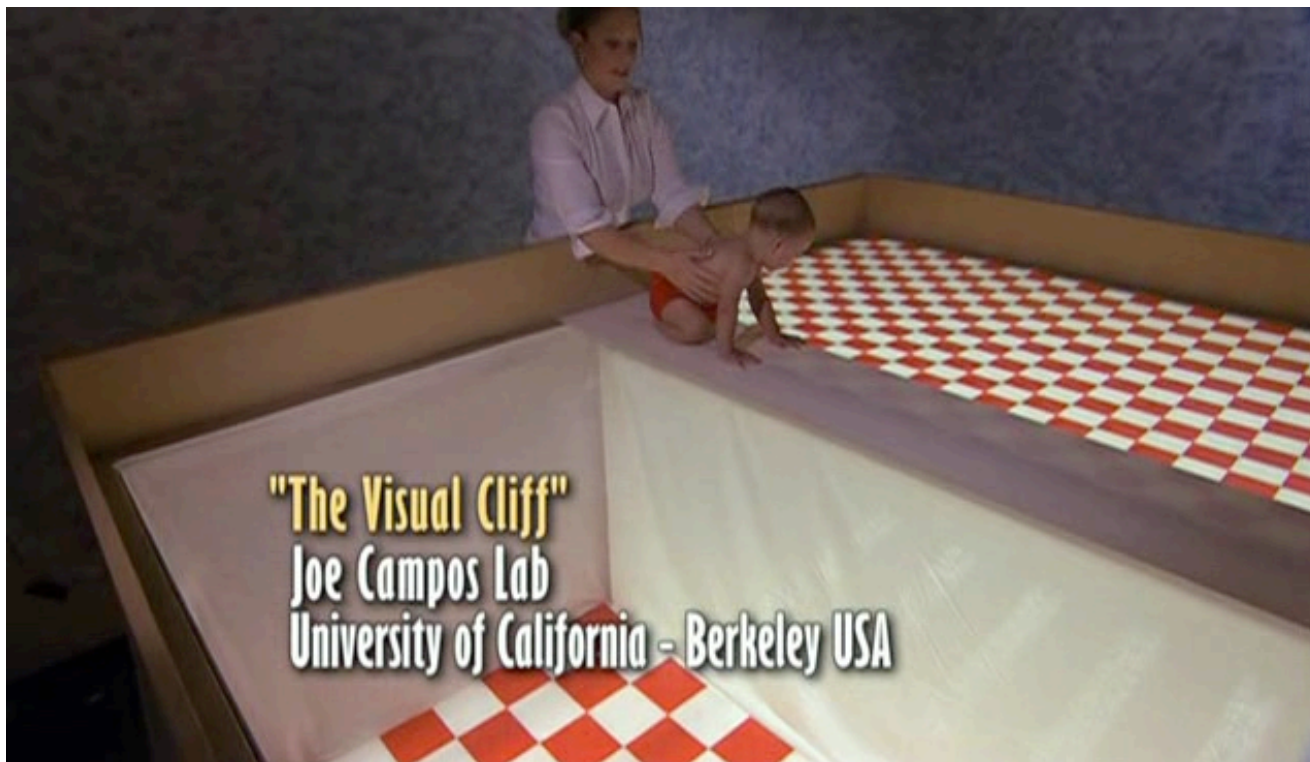


FIGURE 1 Example of an experimental display presented in Experiment 1. Both texture gradients and relative height in the visual field specify the lower hippo to be apparently nearer to the observer.



Auditory development

Auditory development

- 1 month: infants can't hear very quiet sounds quite as well as adults can
- 6 months: almost adult-like
- School age: adult-like
- Old age: lose high frequencies first...



You can only hear whatever you're reading

Brainstorm vs. Green needle



Speech-to-Song illusion

Public Service Announcement

WEAR EARPLUGS AT LOUD CONCERTS (I like Etymotic earplugs)



Cochlear implants

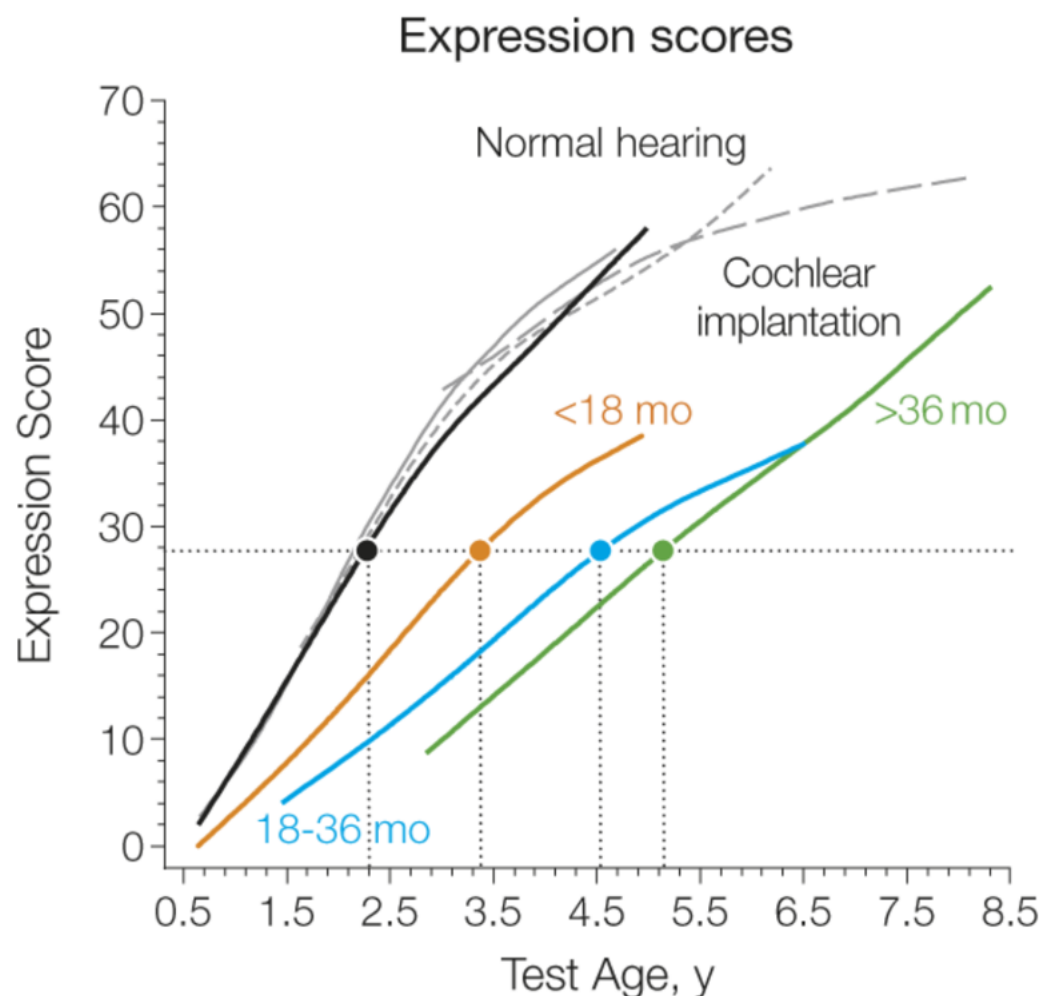






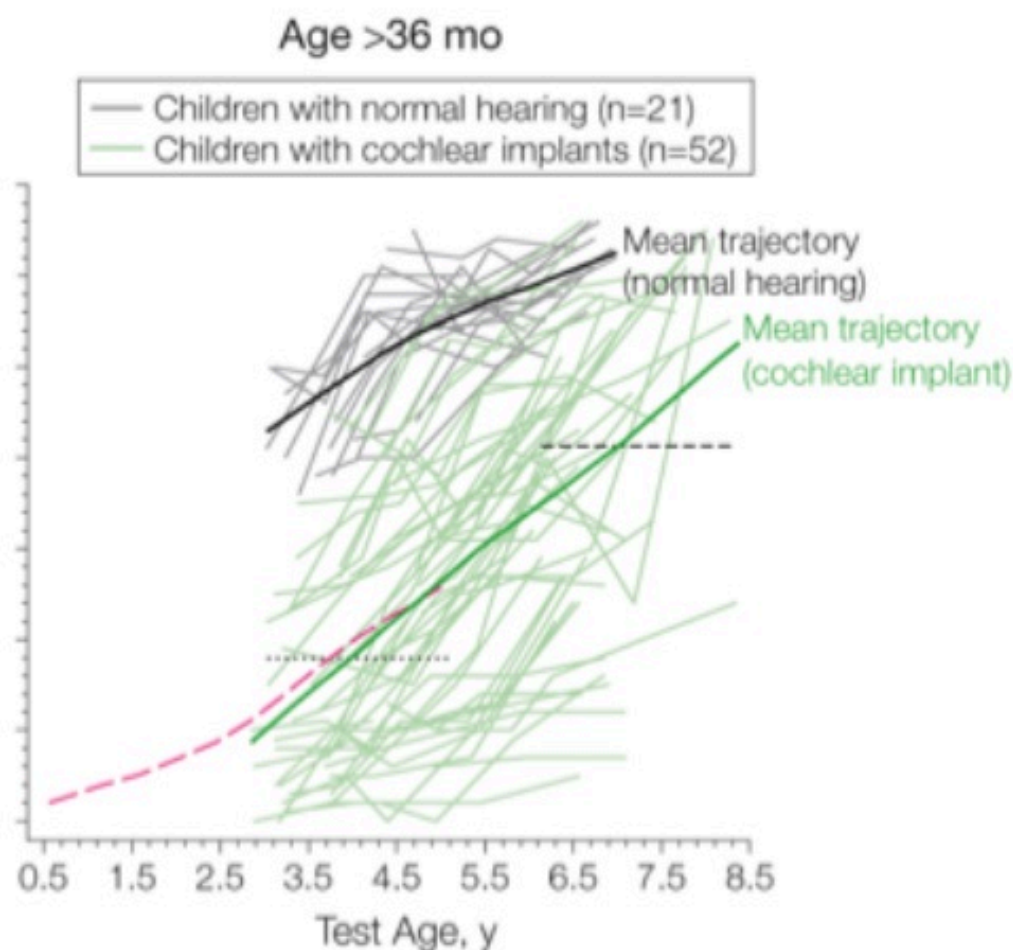
Cochlear implants and language skills

Study of children who received CIs as infants (<18 months), toddlers (18-36 months), or preschoolers (3-5 years)

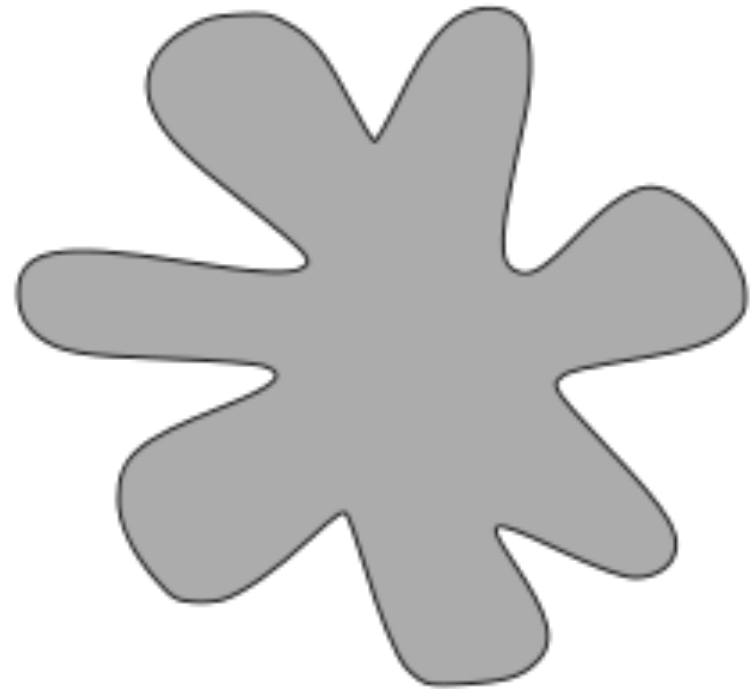
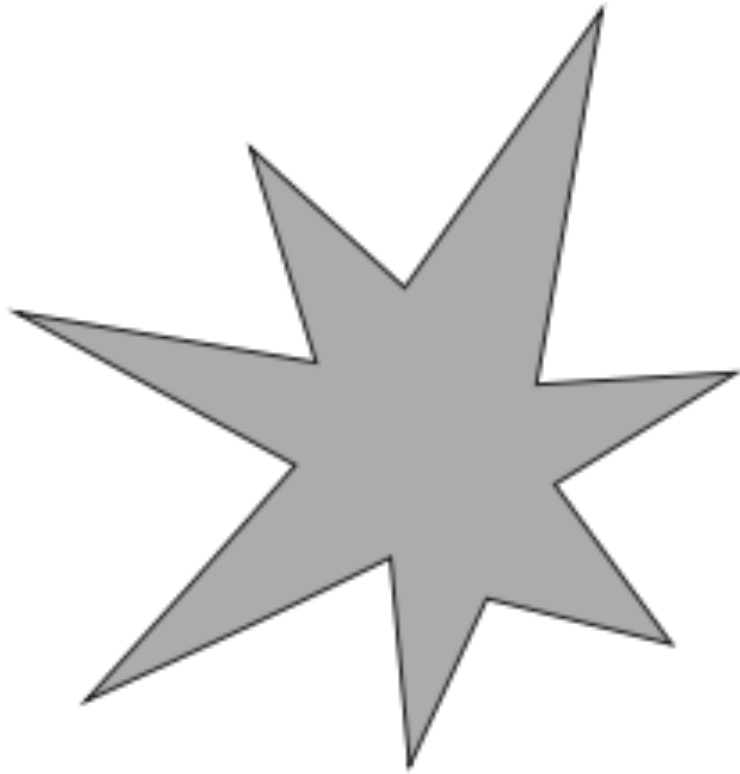


Cochlear implants and language skills

Study of children who received CIs as infants (<18 months), toddlers (18-36 months), or preschoolers (3-5 years)



Intermodal perception



Which of these shapes is **Bouba**?
Which is **Kiki**?